

Question				Marking details	Marks available					
					AO1	AO2	AO3	Maths	Prac	Total
11.	(a)	(i)				1				1
		(ii)		formed from two molecules of the same acid	1					1
	(b)	(i)	I	$^+\text{NH}_3 - \text{CH}_2 - \text{COO}^-$	1					1
			II	it does not have an acidic hydrogen atom (in the 'carboxylic acid' group)			1			1
		(ii)	I	$\text{C}_3\text{H}_5\text{NO}^+ / \text{CH}_2\text{-C(O)-N(H)-CH}_2^+$			1			1
			II	award (1) for two correct absorption values / ranges $\text{N-H} \sim 3000\text{-}3500 \text{ cm}^{-1}$ $\text{C=O} \sim 1650\text{-}1750 \text{ cm}^{-1}$		1				1
	(c)	(i)		0.53		1			1	1
		(ii)		dye absorbs in the yellow / green region colours seen comprises the rest / is the complimentary colour / is made from red and blue			1			2

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		(iii)		$E = \frac{hc}{\lambda} \quad (1)$ $E = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{564 \times 10^{-9}} = 3.53 \times 10^{-19} \text{ J} \quad (1)$ $E = \frac{3.53 \times 10^{-19} \times 6.02 \times 10^{23}}{1000} = 212 \text{ kJ mol}^{-1} \quad (1)$	1	1	1	2		3
	(d)	(i)		M_r of 2-aminohexanoic acid = 131 (1) moles of 2-aminohexanoic acid = $\frac{0.500}{131} = 3.82 \times 10^{-3} \text{ mol}$ gives $3.82 \times 10^{-3} \text{ mol}$ of nitrogen volume of nitrogen = $3.82 \times 10^{-3} \times 24.5 \times 1000 = 93.5 \text{ cm}^3 \quad (1)$ ecf possible e.g. from incorrect M_r	1	1		1	1	2
		(ii)		award (1) each for any two of following 2-aminohexanoic acid is impure temperature less than 298K / pressure greater than 1 atm incomplete reaction nitrogen is lost side reactions		1	1		2	2
				Total question 11	4	6	5	3	4	15